

Invoixy — Interactive Presentation Guide

Live Demonstration Script • Architectural Specs • Examiner Viva Defense

🌐 Production URL: <https://invoixy.vercel.app> (Click to Open)

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🗨️ PRESENTER PRO-TIP FOR TOMORROW MORNING:

Open the live app at <https://invoixy.vercel.app> in one browser tab, and keep this interactive PDF open side-by-side. You can click the links inside this PDF to trigger the exact pages during your demonstration!

Step-by-Step Presentation Walkthrough

Read the speech boxes aloud while executing the action boxes on screen.

🎤 STEP 1: INTRODUCTION & THE 3 BILLING HEADACHES (30 SECONDS)

Respected professors and evaluators, today I am presenting Invoixy, a modern cloud-based GST Invoicing platform designed for retail and wholesale businesses. In traditional retail, shopkeepers face three major problems: manual math errors calculating taxes, long customer wait lines, and strict Rule 46 GST compliance. Invoixy automates the entire process, generating compliant bills in under 30 seconds.

👉 ON-SCREEN ACTION: OPEN LOGIN SCREEN & MOUNT DEMO SANDBOX

Point to the login options (Google OAuth and Email). Then click 'Want to test first? Sign in as Demo !' to instantly mount the multi-device demo store.

👉 Click here to test live: <https://invoixy.vercel.app/login>

🎤 STEP 2: BUSINESS CONTEXT — APEX ELECTRONICS & APPLIANCES

To demonstrate real-world commercial viability, we configured Invoixy for Apex Electronics & Appliances in Kurla West, Mumbai. Electronics retail is the ideal showcase because items span multiple GST slabs—from 18% on laptops and TVs to 28% on air conditioners—with mandatory HSN codes for manufacturer warranty claims.

👉 ON-SCREEN ACTION: SHOW STORE DASHBOARD & REVENUE METRICS

Highlight the summary metric cards: Total Revenue, Paid Invoices count, Pending Bills, and the Recent Invoices ledger.

👉 Click here to test live: <https://invoixy.vercel.app/>

The Core Showcase: Creating an Invoice in 30s

👉 ON-SCREEN ACTION: NAVIGATE TO NEW INVOICE WORKSPACE

Click 'New Invoice' in the top navbar. Point out the split layout: Data Entry Form on the left and Live Interactive A4 Preview on the right.

👉 Click here to test live: <https://invoixy.vercel.app/invoices/new>

👉 STEP 3: EXPLAINING 1-CLICK CUSTOMER & PRODUCT PRESETS

Watch how fast billing is. Instead of typing addresses and tax numbers manually, I select 'Rahul Sharma' from Saved Customers. His Mumbai address, GSTIN, and Maharashtra state code (27) auto-fill instantly. Next, I pick 'Sony Bravia 55-inch 4K TV' from Saved Products. The description, HSN code 8528, unit, and 18% tax rate load in one click.

👉 STEP 4: THE LIVE GST CALCULATION & INTERACTIVE A4 PREVIEW

Behind the scenes, Invoixy compares our seller state code (27) with the buyer's state code (27). Because they match, it automatically classifies this as Intra-State and divides the 18% tax into 9% CGST and 9% SGST. The nearest-rupee round-off is calculated automatically. On the right, the live A4 preview reflects every change instantly.

👉 ON-SCREEN ACTION: PRINT SERVER-SIDE VECTOR PDF

Click the 'Print / PDF' button. Show the official Tax Invoice featuring the high-res Apex logo, itemized table, bank remittance details, and total in Indian words.

👉 Click here to test live: <https://invoixy.vercel.app/invoices/new>

Indian GST Engine Mathematical Formulation

Mathematical logic implemented in `src/lib/gst.ts` following CGST Rule 46.

1. Taxable Amount Calculation

`TaxableAmount = Quantity * Rate * (1 - (DiscountPercent / 100))`

2. Intra-State Tax Split (Seller Code == Buyer Code == 27)

`CGSTRate = GSTRate / 2 | SGSTRate = GSTRate / 2 | IGSTRate = 0`
`CGSTAmount = TaxableAmount * (CGSTRate / 100)`
`SGSTAmount = TaxableAmount * (SGSTRate / 100)`

3. Inter-State Tax Logic (Seller Code != Buyer Code)

`IGSTRate = GSTRate | CGSTRate = 0 | SGSTRate = 0`
`IGSTAmount = TaxableAmount * (IGSTRate / 100)`

4. Nearest-Rupee Round-Off Formulation

`UnroundedTotal = Subtotal + TotalTax`
`GrandTotal = Math.round(UnroundedTotal)`
`RoundOff = GrandTotal - UnroundedTotal`

5. Indian Number-to-Words Algorithm

Converts totals into formal Indian currency words (Lakhs and Crores):
Example: Rs. 2,35,888 -> 'INR Two Lakh Thirty-Five Thousand Eight Hundred Eighty-Eight Only'

Full-Stack Architecture & 10-Model Schema

Next.js 16 Serverless + Prisma ORM + Turso Cloud Database (AWS ap-south-1).

- 1. **User:** Authentication identity (Google OAuth, name, email, avatar).
- 2. **Shop:** Multi-tenant business container (Apex Electronics, Demo Shop).
- 3. **ShopMember:** RBAC junction table linking User to Shop with roles (OWNER, MEMBER).
- 4. **SellerSettings:** Business identity: GSTIN, PAN, address, bank IFSC, and logo base64.
- 5. **Customer:** Preset buyer address book with state code mapping.
- 6. **Product:** Inventory catalog with HSN/SAC codes, rates, and default tax slabs.
- 7. **Invoice:** Immutable snapshot of issued bill with sequence and payment status.
- 8. **InvoiceLine:** Individual item breakdown with quantity, unit rate, and tax values.
- 9. **InvoiceSequence:** Atomic sequence counter preventing invoice number collisions.
- 10. **SecuritySettings:** Terminal inactivity timer and counter lock credentials.

🔒 MULTI-TENANCY ISOLATION GUARANTEE:

Every query for customers, products, and invoices is filtered strictly by 'where: { shopId: session.activeShopId }'. User A can never access User B's billing records. The demo sandbox operates in its own dedicated, isolated shop container.

Top 10 Viva & Examiner Defense Questions

Memorize or refer to these high-scoring technical answers during your viva.

Q1. What is the architecture of Invoixy?

Full-Stack Serverless Architecture on Next.js 16 with React Server Components (RSC) and Server Actions connected to a Turso distributed edge database.

Q2. Why did you use Turso (LibSQL) instead of MySQL or MongoDB?

Turso is a serverless, distributed SQLite edge database hosted in AWS Mumbai. It offers sub-20ms query latency, zero server maintenance, and native atomic transactions.

Q3. How do you prevent duplicate invoice numbers?

We maintain an InvoiceSequence table inside an atomic database transaction (prisma.\$transaction). The counter is incremented with an atomic lock per financial year.

Q4. How is multi-tenancy enforced?

Every record has a mandatory shopId foreign key. All queries filter strictly by activeShopId resolved from the cryptographically signed JWT session cookie.

Q5. Why server-side PDFKit instead of window.print()?

Browser printing depends on printer drivers, page margins, and zoom levels, which break invoice formatting. PDFKit generates vector-crisp, identical A4 documents on every device.

